

Clarifying the Index Ordering in Gradient and Divergence Operations

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The Issue: Gradient and Divergence Index Conventions

In continuum mechanics, there is a **persistent notational discrepancy** in how the gradient of a vector and the divergence of a tensor are written. This arises from:

1. The **natural mathematical definition** of the gradient operator. 2. The **comma notation** for partial derivatives, which can flip indices unintentionally.

Correct Definitions

Gradient of a Vector

The gradient of a vector $\mathbf{v}$ should form a proper second-order tensor:

$$\nabla \mathbf{v} = \frac{\partial v_j}{\partial x_i} \mathbf{e}_i \otimes \mathbf{e}_j = v_{j,i} \mathbf{e}_i \otimes \mathbf{e}_j,$$

where:

- i = **gradient index** (row index, associated with ∇).
- j = **vector component index** (column index).

Divergence of a Tensor

The divergence contracts the **gradient index** (first index):

$$\nabla \cdot \mathbf{S} = \frac{\partial S_{ji}}{\partial x_j} \mathbf{e}_i = S_{ji,j} \mathbf{e}_i.$$

Common Mistake

Many texts incorrectly write:

$$\nabla \mathbf{v} = v_{i,j} \mathbf{e}_i \otimes \mathbf{e}_j \quad (\text{transposed!})$$

This leads to the **wrong divergence**:

$$\nabla \cdot \mathbf{S} = S_{ij,j} \mathbf{e}_i \quad (\text{only valid if } \mathbf{S} \text{ is symmetric}).$$

Why This Matters

- If $\mathbf{S}$ is **non-symmetric** (e.g., in polar materials), $S_{ij,j}$ is **incorrect**.
- The comma notation $v_{i,j}$ **hides the transposition** inherent in $\nabla \mathbf{v}$.

Recommended References

- **Gurtin et al. (2010)**: Uses $v_{j,i}$ and $S_{ji,j}$ consistently.
- **Truesdell & Noll (2004)**: Explicitly warns about index ordering.
- **Holzpfel (2000)**: Correctly uses $\sigma_{ji,j}$ for divergence.

Key Takeaway

Always:

$$\nabla \mathbf{v} = v_{j,i} \mathbf{e}_i \otimes \mathbf{e}_j, \quad \nabla \cdot \mathbf{S} = S_{ji,j} \mathbf{e}_i.$$

Verify if authors assume symmetry ($S_{ij} = S_{ji}$) before using $S_{ij,j}$.